

outside consultation. We implicitly promote the ivory tower.”

Vikram Sarabhai

Scientist 1912-1971

MASTER MIND

With an astonishing diversity of interests, the son of a rich textile mill owner wrote 80-odd scientific papers, set up 40 institutions, initiated the nation's space programme and headed its nuclear mission in the '60s

■ By Amrita Shah

At the age of 11 or so, Vikram Sarabhai was said to have been fond of setting a challenge for himself: riding a bicycle which he would pedal as fast as he could. When it had achieved a satisfactory momentum, he would cross his arms over his chest and place his feet on the handlebars, close his eyes if the road was straight. This way he would let the bicycle carry him as far as it would go, impervious to the pleas of his terrified servants chasing him and begging him to stop.

It is an odd bit of daredevilry to associate with a man credited with writing 80-odd scientific papers, setting up almost 40 institutions, initiating India's space programme and heading its nuclear energy programme in the '60s. On the other hand, in that boyhood act of heroism one can probably find the key, the essential mix of daring, skill and audacity that made his life extraordinary.

The seeds of this remarkable life were probably sown in early childhood and fostered by his unusual upbringing. Sarabhai came from a prosperous Jain family of textile mill owners in Ahmedabad. The Sarabhais, in addition to being wealthy, were close to Mahatma Gandhi and were known for their strong sense of social responsibility: Vikram's aunt, Ansuya, founded the city's first trade union of textile workers and his sister Mridula was actively involved in the freedom movement

and went to jail many times.

Vikram and his seven siblings were educated in a private experimental school on the family's 21-acre property, an education that included a range of extra-curricular activities as well as an exposure to distinguished visitors such as Rabindranath Tagore, Jawaharlal Nehru and Rukmini Devi Arundale. He went on to pursue physics at the Indian Institute of Science in Bangalore—where he studied under Nobel laureate C.V. Raman and struck a friendship with Homi Bhabha, who later set up India's atomic energy programme—and later at Cambridge.

In the fervently idealistic post-Independence era, Sarabhai established several institutions such as the Physical Research Laboratory; the Darpana Dance Academy, which he co-founded with his wife, Mrinalini; the Ahmedabad Textile Industry's Research Association (ATIRA), India's first textile research cooperative; the country's first market research agency, the Operations Research Group; the Indian Institute of Management (Ahmedabad) and helped in setting up the National Institute of Design. The range of activities reveals the astonishing diversity of his interests and also the consistency of his approach which involved the use of scientific methods, sound financial plan and a clear nationalistic purpose.

All these features were evident in his conceptualisation of the space programme. In the early '60s when space

technology was being used for military purposes and one-upmanship between the superpowers, the idea of a poor struggling nation aspiring to a space programme of its own seemed a fairly unthinkable one. Sarabhai made it seem not only feasible but essential by setting targets for the development of technology and mapping out a range of applications—communications, weather forecasting and mineral prospecting—that could answer the needs of a developing nation.

All his projects, whether it was the satellite instructional television experiment which in 1975-76 used a NASA satellite to beam educational content to 2,400 of India's poorest villages, or his plan for building agricultural complexes serviced by atomic power and desalinated sea water, were fired with an imaginative zeal. He worked 18-20 hours every day using both charm and persuasion to convince others, including his seniors like Bhabha, and industrialist Kasturbhai Lalbhai, who collaborated in many of his ventures, and his subordinates to share his dreams. He did have his share of critics. There were some who found his ideas wildly impractical. And others who did not appreciate his complex and highly nuanced approach to the sensitive issue of nuclear weapons. It is hard to dispute though, that when he passed away, on December 30, 1971, at the age of 52, India lost one of her most dynamic and endearing visionaries.

The author has written Sarabhai's biography, Vikram Sarabhai: A Life